

Ex. No:6 Use Sleuth Kit to analyze digital evidence

Date :

Description

The Sleuth Kit (TSK) is a collection of command-line tools that allow you to analyze disk images and recover digital evidence. Here's a step-by-step guide to using Sleuth Kit on a Windows machine to analyze digital evidence:

Step 1: Install Sleuth Kit

1. Download Sleuth Kit:

- Visit the official [Sleuth Kit website](https://www.sleuthkit.org/) or https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v
- and download the latest version for Windows.

2. Install Sleuth Kit:

- Run the installer and follow the instructions to install Sleuth Kit on your Windows machine.

Step 2: Acquire the Disk Image

Before analysis, you need a disk image of the evidence. This can be an image of a hard drive, memory card, or any other storage device.

1. Create Disk Image:

- Use a tool like FTK Imager or dd to create a bit-by-bit copy of the storage device. Ensure the image is in a format supported by Sleuth Kit, such as .dd, .raw, .img, or .E01.
- Download the below files from the google drive
 1. 4Dell Latitude CPi.E01
 2. 4Dell Latitude CPi.E02

Step 3: Mount the Disk Image (Optional)

Mounting the disk image makes it easier to analyze the file system.

1. Mount the Image:

- You can use a tool like OSFMount to mount the image as a virtual drive on your Windows system.
- This step is optional but helps with navigating the file system easily.

Step 4: Analyze the File System

Use Sleuth Kit tools to analyze the file system and locate evidence.

1. Navigate to the Sleuth Kit Directory:

- Open the Command Prompt and navigate to the directory where Sleuth Kit is installed.

```
C:\Users\Admin\Pictures\sleuthkit-4.14.0-win32\bin>mmls "%USERPROFILE%\Pictures\4Dell Latitude CPi.E01"
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors
```

	Slot	Start	End	Length	Description
000:	Meta	0000000000	0000000000	0000000001	Primary Table (#0)
001:	-----	0000000000	0000000062	0000000063	Unallocated
002:	000:000	0000000063	0009510479	0009510417	NTFS / exFAT (0x07)
003:	-----	0009510480	0009514259	0000003780	Unallocated

```
C:\Users\Admin\Pictures\sleuthkit-4.14.0-win32\bin>
```

2. Identify File System Type with fsstat:

- Run the command:

arduino

fsstat [image file] > filesystem_info.txt

- This command outputs information about the file system, which is crucial for understanding the structure of the disk.

```
C:\Users\Admin\Pictures\sleuthkit-4.14.0-win32\bin>fsstat -o 63 "%USERPROFILE%\Pictures\4Dell Latitude CPi.E01"
FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: B26CB1CE6CB18D9B
OEM Name: NTFS
Version: Windows XP

METADATA INFORMATION
-----
First Cluster of MFT: 2097152
First Cluster of MFT Mirror: 4755208
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 12305
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 0 - 9510415
Total Sector Range: 0 - 9510415

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16) Size: 48-72 Flags: Resident
$ATTRIBUTE_LIST (32) Size: No Limit Flags: Non-resident
$FILE_NAME (48) Size: 68-578 Flags: Resident,Index
$OBJECT_ID (64) Size: 0-256 Flags: Resident
$SECURITY_DESCRIPTOR (80) Size: No Limit Flags: Non-resident
$VOLUME_NAME (96) Size: 2-256 Flags: Resident
$VOLUME_INFORMATION (112) Size: 12-12 Flags: Resident
$DATA (128) Size: No Limit Flags:
$INDEX_ROOT (144) Size: No Limit Flags: Resident
$INDEX_ALLOCATION (160) Size: No Limit Flags: Non-resident
$BITMAP (176) Size: No Limit Flags: Non-resident
$REPARSE_POINT (192) Size: 0-16384 Flags: Non-resident
$EA_INFORMATION (208) Size: 8-8 Flags: Resident
$EA (224) Size: 0-65536 Flags:
$LOGGED_UTILITY_STREAM (256) Size: 0-65536 Flags: Non-resident

C:\Users\Admin\Pictures\sleuthkit-4.14.0-win32\bin>
```

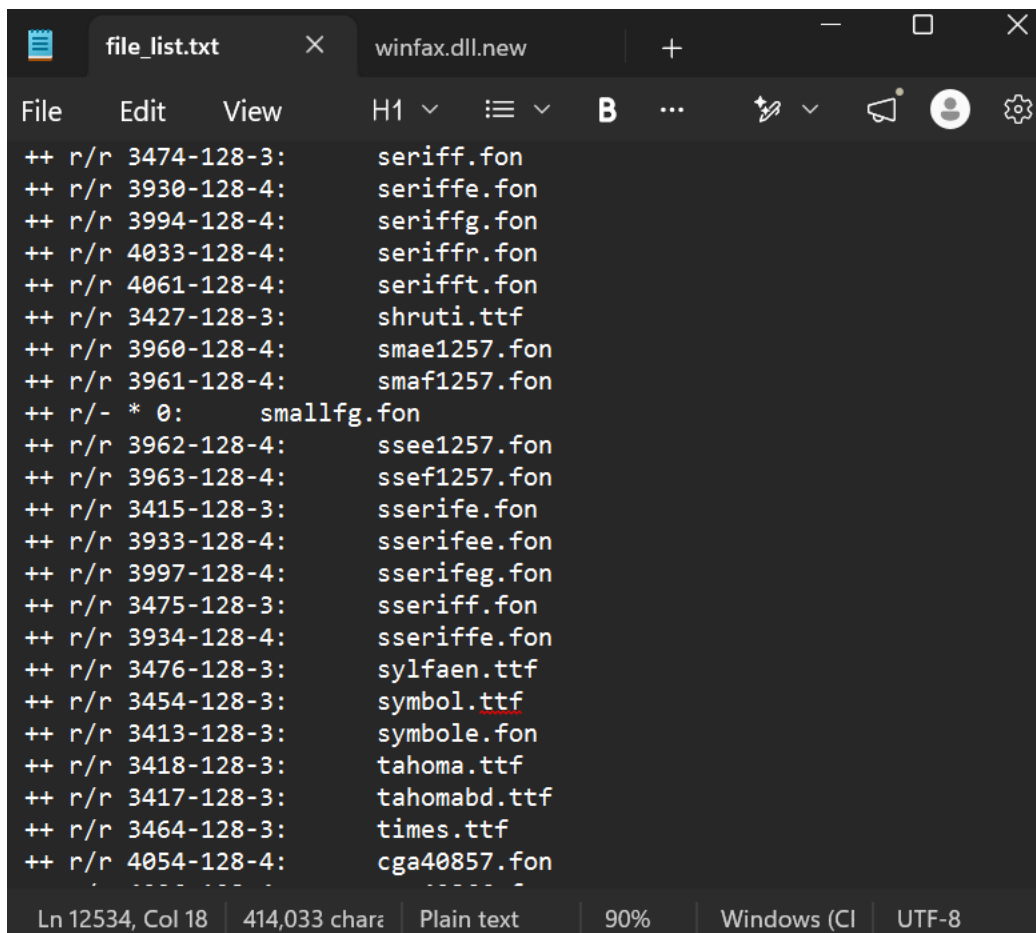
3. List Partitions with mmls:

- Run the command:

arduino

mmls [image file] > partitions.txt

- This command lists the partitions within the image file.



```
file_list.txt
winfax.dll.new
File Edit View H1  B ...
++ r/r 3474-128-3: serif.fon
++ r/r 3930-128-4: seriffe.fon
++ r/r 3994-128-4: seriffg.fon
++ r/r 4033-128-4: seriffr.fon
++ r/r 4061-128-4: serifft.fon
++ r/r 3427-128-3: shruti.ttf
++ r/r 3960-128-4: smae1257.fon
++ r/r 3961-128-4: smaf1257.fon
++ r/- * 0: smallfg.fon
++ r/r 3962-128-4: ssee1257.fon
++ r/r 3963-128-4: ssef1257.fon
++ r/r 3415-128-3: sserife.fon
++ r/r 3933-128-4: sserifee.fon
++ r/r 3997-128-4: sserifeg.fon
++ r/r 3475-128-3: sseriff.fon
++ r/r 3934-128-4: sseriffe.fon
++ r/r 3476-128-3: sylfaen.ttf
++ r/r 3454-128-3: symbol.ttf
++ r/r 3413-128-3: symbole.fon
++ r/r 3418-128-3: tahoma.ttf
++ r/r 3417-128-3: tahomabd.ttf
++ r/r 3464-128-3: times.ttf
++ r/r 4054-128-4: cga40857.fon
Ln 12534, Col 18 | 414,033 chara | Plain text | 90% | Windows (CI | UTF-8
```

4. Analyze File System with fls:

- Run the command:

arduino

```
fls -r [image file] > file_list.txt
```

1. This command recursively lists files and directories in the file system, showing their metadata.

5. Recover Deleted Files with icat:

- To extract a specific file, use icat:

css

```
icat [image file] [inode number] > [output file]
```

- Replace [inode number] with the inode of the file you want to recover, which you can find from the fls output.

Step 5: Analyze Metadata

Extract metadata from files to understand more about the file's history.

1. **View Metadata with istat:**

- Run the command:

This provides detailed information about a file, including timestamps, size, and allocation status

```
C:\Users\Admin\Pictures\sleuthkit-4.14.0-win32\bin>fls -o 63 -r -d "%USERPROFILE%\Pictures\4Dell Latitude Cpi.E01"
r/- * 0: Documents and Settings/Default User/MPC7A4.tmp
r/- * 0: Documents and Settings/Default User/MPC7A5.tmp
r/- * 6366: Documents and Settings/Default User/MPC7A6.tmp
r/- * 6367: Documents and Settings/Default User/MPC7A7.tmp
r/- * 6368: Documents and Settings/Default User/MPC7A8.tmp
r/- * 6369: Documents and Settings/Default User/MPC7A9.tmp
r/- * 6370: Documents and Settings/Default User/MPC7AA.tmp
r/- * 6371: Documents and Settings/Default User/MPC7AB.tmp
r/- * 6372: Documents and Settings/Default User/MPC7AC.tmp
r/- * 0: Documents and Settings/Default User/MPC7B8.tmp
r/- * 0: Documents and Settings/Default User/MPC7B9.tmp
r/- * 6386: Documents and Settings/Default User/MPC7BA.tmp
r/- * 6387: Documents and Settings/Default User/MPC7BB.tmp
r/- * 6388: Documents and Settings/Default User/MPC7BC.tmp
r/- * 6389: Documents and Settings/Default User/MPC7BD.tmp
r/- * 6390: Documents and Settings/Default User/MPC7BE.tmp
r/- * 6391: Documents and Settings/Default User/MPC7BF.tmp
r/- * 6392: Documents and Settings/Default User/MPC7C0.tmp
r/- * 0: Documents and Settings/Default User/MPC7CC.tmp
r/- * 0: Documents and Settings/Default User/MPC7CE.tmp
r/- * 6408: Documents and Settings/Default User/MPC7CF.tmp
r/- * 6409: Documents and Settings/Default User/MPC7D0.tmp
r/- * 6410: Documents and Settings/Default User/MPC7D1.tmp
r/- * 6411: Documents and Settings/Default User/MPC7D2.tmp
r/- * 6412: Documents and Settings/Default User/MPC7D3.tmp
r/- * 6413: Documents and Settings/Default User/MPC7D4.tmp
r/- * 0: Documents and Settings/Default User/MPC7E0.tmp
r/- * 0: Documents and Settings/Default User/MPC7E2.tmp
r/- * 6428: Documents and Settings/Default User/MPC7E3.tmp
r/- * 6429: Documents and Settings/Default User/MPC7E4.tmp
r/- * 6430: Documents and Settings/Default User/MPC7E5.tmp
r/- * 6431: Documents and Settings/Default User/MPC7E6.tmp
r/- * 6432: Documents and Settings/Default User/MPC7E7.tmp
r/- * 6433: Documents and Settings/Default User/MPC7E8.tmp
r/- * 0: Documents and Settings/Default User/MPC7F4.tmp
r/- * 0: Documents and Settings/Default User/MPC7F6.tmp
r/- * 6448: Documents and Settings/Default User/MPC7F7.tmp
r/- * 6449: Documents and Settings/Default User/MPC7F8.tmp
```

CSS

istat [image file] [inode number] > metadata_info.txt

.

Step 6: Timeline Analysis (Optional)

Creating a timeline of file activity can be crucial in an investigation.

1. Create Timeline with mactime:

- Generate a body file using fls:

CSS

```
C:\Users\Admin\Pictures\sleuthkit-4.14.0-win32\bin>istat -o 63 "%USERPROFILE%\Pictures\4Dell Latitude Cpi.E01" 9522
MFT Entry Header Values:
Entry: 9522 Sequence: 1
$LogFile Sequence Number: 25787708
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Archive, Compressed
Owner ID: 0
Security ID: 0 ()
Created: 2001-08-23 23:30:00.000000000 (India Standard Time)
File Modified: 2001-08-23 23:30:00.000000000 (India Standard Time)
MFT Modified: 2004-08-20 04:18:03.705616000 (India Standard Time)
Accessed: 2004-08-20 04:18:03.705616000 (India Standard Time)

$FILE_NAME Attribute Values:
Flags: Archive, Compressed
Name: winfax.dll
Parent MFT Entry: 688 Sequence: 688
Allocated Size: 16384 Actual Size: 9216
Created: 2001-08-23 23:30:00.000000000 (India Standard Time)
File Modified: 2001-08-23 23:30:00.000000000 (India Standard Time)
MFT Modified: 2004-08-19 22:32:20.051385600 (India Standard Time)
Accessed: 2004-08-20 04:18:03.705616000 (India Standard Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0) Name: N/A Resident size: 72
Type: $FILE_NAME (48-5) Name: N/A Resident size: 86
Type: $SECURITY_DESCRIPTOR (80-4) Name: N/A Resident size: 100
Type: $DATA (128-1) Name: N/A Non-Resident, Compressed size: 9216 init_size: 9216
3344644 3344645 3344646 3344647 3344648 3344649 3344650 3344651
3344652 0 0 0 0 0 0
3344653 0
```

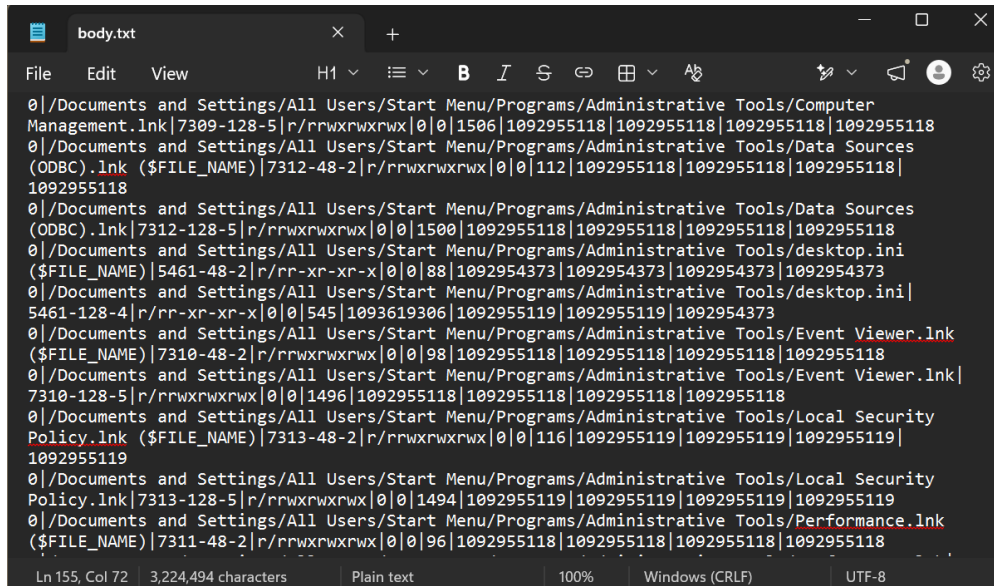
fls -m / -r [image file] > body.txt

- Then create the timeline:

css

mactime -b body.txt > timeline.txt

- The timeline includes MAC (Modified, Accessed, Changed) times of files.



```
File Edit View H1 i B I S A
0|/Documents and Settings/All Users/Start Menu/Programs/Administrative Tools/Computer
Management.lnk|7309-128-5|r/rrwxrwxrwx|0|0|1506|1092955118|1092955118|1092955118|1092955118
0|/Documents and Settings/All Users/Start Menu/Programs/Administrative Tools/Data Sources
(ODBC).lnk ($FILE_NAME)|7312-48-2|r/rrwxrwxrwx|0|0|112|1092955118|1092955118|1092955118|
1092955118
0|/Documents and Settings/All Users/Start Menu/Programs/Administrative Tools/Data Sources
(ODBC).lnk|7312-128-5|r/rrwxrwxrwx|0|0|1500|1092955118|1092955118|1092955118|1092955118
0|/Documents and Settings/All Users/Start Menu/Programs/Administrative Tools/desktop.ini
($FILE_NAME)|5461-48-2|r/rr-xr-xr-x|0|0|88|1092954373|1092954373|1092954373|1092954373
0|/Documents and Settings/All Users/Start Menu/Programs/Administrative Tools/desktop.ini|
5461-128-4|r/rr-xr-xr-x|0|0|545|1093619306|1092955119|1092955119|1092954373
0|/Documents and Settings/All Users/Start Menu/Programs/Administrative Tools/Event Viewer.lnk
($FILE_NAME)|7310-48-2|r/rrwxrwxrwx|0|0|98|1092955118|1092955118|1092955118|1092955118
0|/Documents and Settings/All Users/Start Menu/Programs/Administrative Tools/Event Viewer.lnk|
7310-128-5|r/rrwxrwxrwx|0|0|1496|1092955118|1092955118|1092955118|1092955118
0|/Documents and Settings/All Users/Start Menu/Programs/Administrative Tools/Local Security
Policy.lnk ($FILE_NAME)|7313-48-2|r/rrwxrwxrwx|0|0|116|1092955119|1092955119|1092955119|
1092955119
0|/Documents and Settings/All Users/Start Menu/Programs/Administrative Tools/Local Security
Policy.lnk|7313-128-5|r/rrwxrwxrwx|0|0|1494|1092955119|1092955119|1092955119|1092955119
0|/Documents and Settings/All Users/Start Menu/Programs/Administrative Tools/Performance.lnk
($FILE_NAME)|7311-48-2|r/rrwxrwxrwx|0|0|96|1092955118|1092955118|1092955118|1092955118
Ln 155, Col 72 | 3,224,494 characters | Plain text | 100% | Windows (CRLF) | UTF-8
```

Step 7: Generate a Report

Document your findings by compiling all the data into a comprehensive report.

1. Compile the Data:

- Gather all the output files (e.g., filesystem_info.txt, partitions.txt, file_list.txt, metadata_info.txt, timeline.txt).

2. Analyze and Document:

- Review the findings, highlight important evidence, and write a report summarizing the investigation's results.

Step 8: Finalize and Store Evidence

Ensure that all evidence and reports are securely stored.

1. Archive Evidence:

- Use a secure method to archive the disk image and analysis results, ensuring integrity and confidentiality.

2. Store Securely:

- Store the archived data in a secure location, following the chain of custody procedures.

By following these steps, you can use Sleuth Kit on a Windows machine to effectively analyze digital evidence and extract crucial information for your investigation.

Rubrics

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	